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**RAFFLES INSTITUTION**
**2011 Year 6 PRELIMINARY EXAMINATION**

**Higher 1**

CANDIDATE NAME

CLASS INDEX NUMBER

# CHEMISTRY 8872/02

Paper 2 **19 September 2011**
**2 hours**

Candidates answer Section A on the Question Paper

Additional Materials: Answer paper
Graph paper
Data Booklet

**READ THESE INSTRUCTIONS FIRST**

**DO NOT open this booklet until you are told to do so.**

Write your name, class and index number in the spaces provided above.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

**Section A (40 marks)**
Answer **all** the questions.

**Section B (40 marks)**
Answer any **two** questions on separate answer paper.

You are to begin each question on a fresh sheet of paper.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets **[ ]** at the end of each question or part question.

You are reminded of the need for good English and clear presentation in your answers.

| For Examiner's Use | |
|---|---|
| **Paper 1** | / 30 |
| **Paper 2** | |
| **A1** | / 8 |
| **A2** | / 10 |
| **A3** | / 12 |
| **A4** | / 10 |
| **B5** | / 20 |
| **B6** | / 20 |
| **B7** | / 20 |
| **Total** | /110 |

## Section A (40 marks)

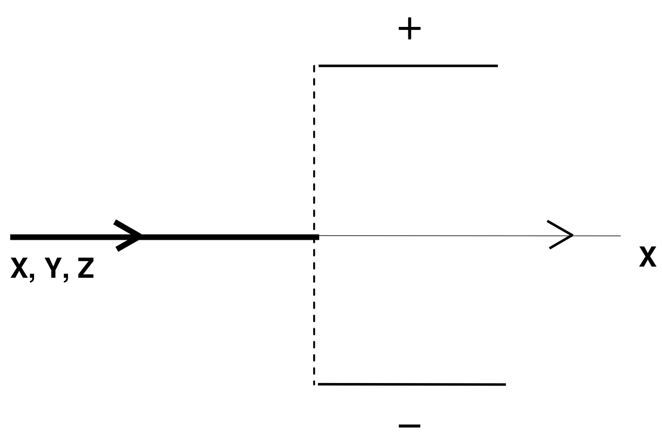
Answer **all** questions in the spaces provided.

**1** **(a)** Complete the table below.

| Particle | Electric charge | Mass number | Number of | | |
|---|---|---|---|---|---|
| | | | Protons | Electrons | Neutrons |
| **X** | 0 | 32 | 16 | 16 | 16 |
| **Y** | 1– | 81 | | 36 | |
| **Z** | | 70 | 31 | 28 | |

**[2]**

**(b)** Beams consisting of particles **X**, **Y** and **Z** are subjected to an electric field as shown in the diagram below. Sketch, and label, on the diagram below to show how beams of **Y** and **Z** are affected by the electric field.




**[2]**

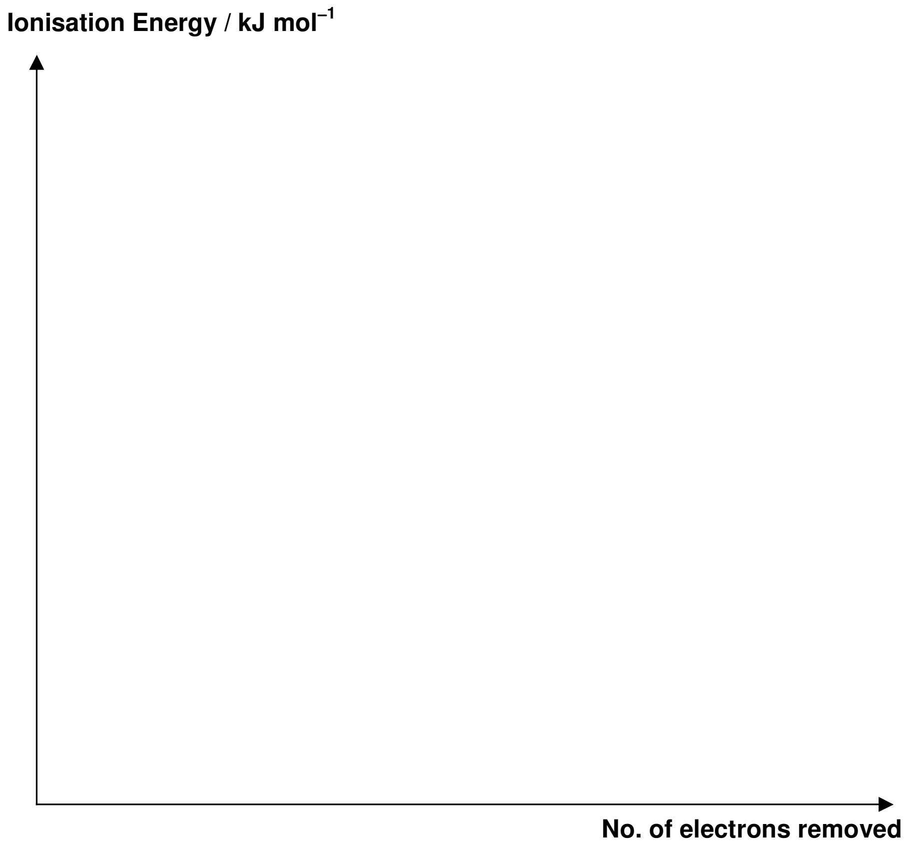
**(c)** Write the electronic configuration for

**(i)** particle **X**

**(ii)** particle **Z**

**[2]**

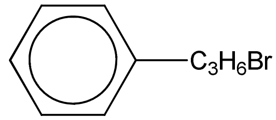
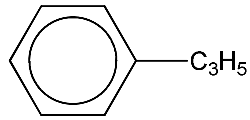
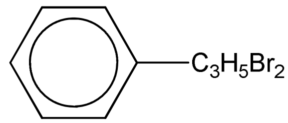
**(d)** Using the grid provided below, sketch a graph of the seven successive ionisation energies for particle **X**.



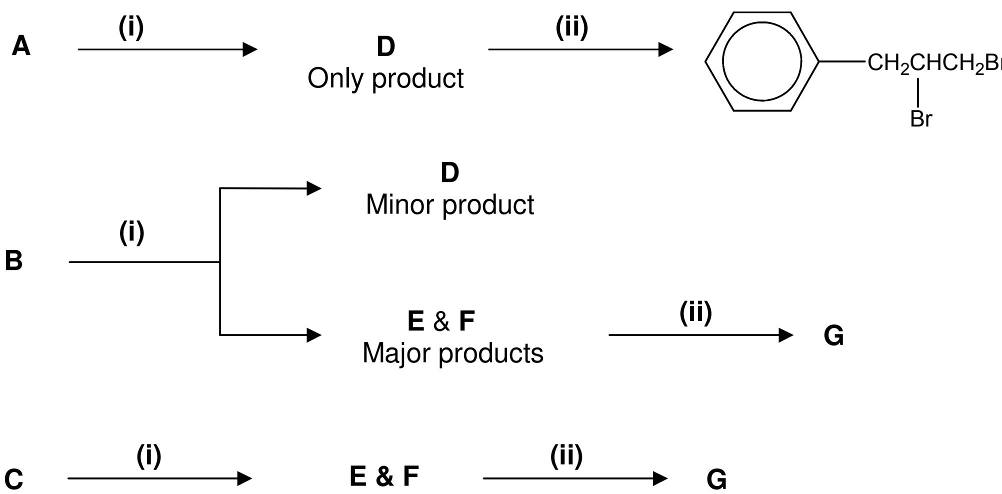

**[2]**

**[Total: 8]**

**2** The partial structural formulae of compounds **A** to **G** are given in the table below.

| Compound | Partial structural formulae |
|---|---|
| **A**, **B**, **C** |  |
| **D**, **E**, **F** |  |
| **G** |  |

The reaction scheme below shows the reactions of **A**, **B**, and **C**.




**(a)** Using the reaction scheme above, complete **Table 1** and **Table 2** below.

**Table 1**

| Step | Reagents and conditions |
|---|---|
| **(i)** | |
| **(ii)** | |

**[2]**

**Table 2: Structural formulae of compounds**

| | |
|---|---|
| **A** | **D** |
| **B** | **E & F** |
| **C** | **G** |

**[6]**

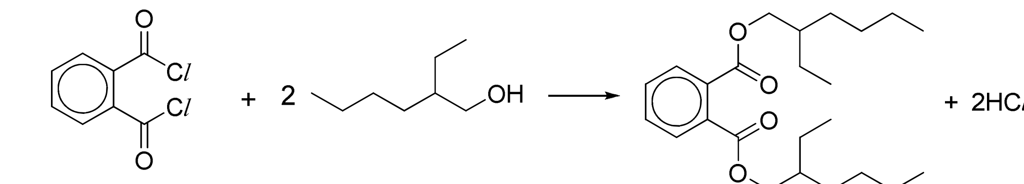
**(b)** State the isomerism exhibited by compounds **E** and **F**, and draw their displayed formulae.

**[2]**

**[Total: 10]**

**3** The 2011 Taiwan food scandal uncovered the illegal use of cancer-causing plasticiser, di(2-ethylhexyl) phthalate or DEHP, as a substitute for palm oil emulsifiers to give sports drinks and fruit juices a “clouded” and “rich” look.

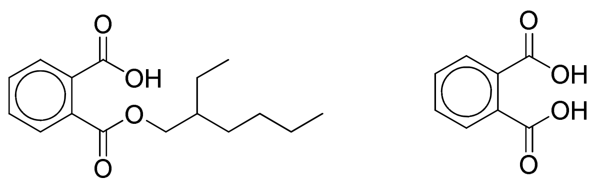
Reports have shown that DEHP is associated with obesity, reduced sperm count in males and cardiotoxicity. DEHP can be synthesized from the reaction of *o*-phthaloyl dichloride with 2-ethylhexanol as shown below:



*o*-phthaloyl dichloride    2-ethylhexanol    DEHP

The DEHP released during the processing of foods can leach into a liquid that comes into contact with the plastic. It is noted that DEHP leaches faster into non-polar solvents (e.g. oils and fats in foods packed in PVC) than polar solvents (e.g. water).

DEHP can be hydrolyzed to MEHP (mono-ethylhexyl phthalate) and subsequently to phthalic acid. The released alcohol, 2-ethylhexanol, is susceptible to oxidation to aldehyde or carboxylic acid.



mono-ethylhexyl phthalate    phthalic acid

**(a)** **(i)** Define the term *bond energy*.

**(ii)** Using bond energy values from the *Data Booklet,* calculate the enthalpy change for the reaction of *o*-phthaloyl dichloride and 2-ethylhexanol to form DEHP.

**[3]**

**(b)** Explain why DEHP leaches faster into non-polar solvents than polar solvents like water.

**[2]**

**(c)** **(i)** State the reagents and conditions for the hydrolysis of DEHP to form phthalic acid and 2-ethylhexanol.

**(ii)** Suggest a simple chemical test to distinguish between 2-ethylhexanol and phthalic acid. State the observations for **each** compound in **each** test.

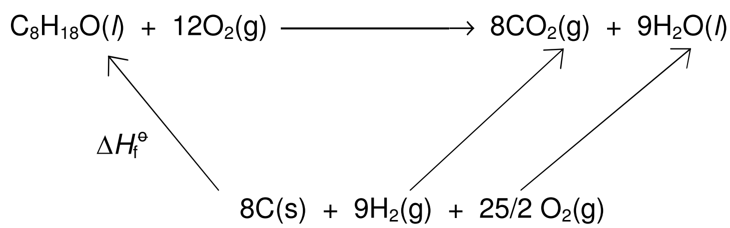
**[3]**

**(d)** In an experiment to determine the enthalpy change of formation of 2-ethylhexanol, 1.5 g of the alcohol was completely burnt in excess oxygen and 57 kJ of heat energy was given off under standard conditions.

**(i)** Calculate the standard enthalpy change of combustion, $\Delta H_c^\ominus$, of 2-ethylhexanol ($C_8H_{18}O$).

**(ii)** Using your answer to **(d)(i)**, together with the $\Delta H$ values and the energy cycle given below, calculate the standard enthalpy change of formation, $\Delta H_f^\ominus$, of 2-ethylhexanol.

$\Delta H_c^\ominus(C) = -393.5$ kJ mol$^{-1}$
$\Delta H_f^\ominus (H_2O) = -285.9$ kJ mol$^{-1}$




**[4]**

**[Total: 12]**

**4** Carbon dioxide, $CO_2$, from burning of fossil fuels, is changing the fundamental chemistry of our oceans. When carbon dioxide reacts with water, it forms carbonic acid, $H_2CO_3$. $H_2CO_3$ subsequently dissociates into bicarbonate ions, $HCO_3^-$, and hydrogen ions, $H^+$, into the sea, lowering pH and causing "acidification" of the ocean. A solution of 0.100 mol dm$^{-3}$ $H_2CO_3$ has a pH of 3.68.

Due to the increasing levels of atmospheric $CO_2$, the oceans have become approximately 30% more acidic (in terms of concentration of $H^+$ ions) over the last 150 years as the pH of seawater decreases from approximately 8.25 to 8.14.

Chemical species like $HCO_3^-$ and $CO_3^{2-}$ are the essential components of the carbonate buffer system which regulates the pH of seawater. The equilibrium reaction for $CO_2$ chemistry in seawater that most cogently captures its behaviour is shown below.

$$CO_2 + CO_3^{2-} + H_2O \rightleftharpoons 2\,HCO_3^-$$

The natural pH of the ocean is determined by a need to balance the deposition and burial of $CaCO_3$ on the sea floor against the influx of $Ca^{2+}$ and $CO_3^{2-}$ into the ocean from weathering of limestone rocks and other minerals on land. Coral reefs are built from limestone, $CaCO_3$, by the reaction shown below.

$$Ca^{2+} + CO_3^{2-} \rightleftharpoons CaCO_3$$

When $CO_2$ increases too rapidly, the natural equilibrium of calcium carbonate is upset. This alters the balance of the buffers and gradually allows the oceans to become more acidic.

Ocean acidification, leading to decrease in shellfish weights and slowed growth of coral reefs, could affect marine food webs and lead to substantial decrease in commercial fish stocks, threatening protein supply and food security for millions of people as well as the multi-billion dollar global fishing industry.

**(a)** **(i)** Explain, with the aid of appropriate calculations, whether carbonic acid, $H_2CO_3$, is a strong or weak acid. You may assume that carbonic acid, $H_2CO_3$, to be monoprotic in your calculations.

**(ii)** Calculate the percentage increase in the concentration of $H^+$ ions in the last 150 years.

**(iii)** With the aid of two balanced equations, show how the carbonate buffer system regulates the pH of the seawater.

**(iv)** Other than increasing levels of atmospheric carbon dioxide gas, suggest another environmental phenomenon that can contribute to ocean acidification.

**[7]**

**(b)** **(i)** State Le Chatelier's principle.

**(ii)** By applying Le Chatelier's principle to these equilibria, explain qualitatively how the rapid increase in carbon dioxide cause the decrease in shellfish weights and slowed growth of coral reefs.

**[3]**

**[Total: 10]**

**Section B (40 marks)**

Answer any **two** questions from this section on separate answer paper.

**5** In 1840, Deville first reportedly prepared dinitrogen pentoxide, $N_2O_5$, an unstable and potentially dangerous oxidant. Dinitrogen pentoxide decomposes into nitrogen dioxide and oxygen gas in tetrachloromethane, $CCl_4$, solution.

$$2N_2O_5(\text{in } CCl_4) \rightleftharpoons 4NO_2(\text{in } CCl_4) + O_2(g)$$

In order to determine the rate equation for this reaction, an investigation was carried out at a constant temperature and measuring the rate of reaction based on decrease in concentration of $N_2O_5$ per second.

The following results were obtained.

| | | | | | | |
|---|---|---|---|---|---|---|
| Rate of reaction / $10^{-5}$ mol dm$^{-3}$ s$^{-1}$ | 0.95 | 1.20 | 1.57 | 1.93 | 2.10 | 2.26 |
| $[N_2O_5]$ / mol dm$^{-3}$ | 0.93 | 1.17 | 1.51 | 1.84 | 2.00 | 2.14 |
| $[N_2O_5]^2$ / mol$^2$ dm$^{-6}$ | 0.86 | 1.37 | 2.28 | 3.39 | 4.00 | 4.58 |

**(a)** **(i)** To determine the order of reaction with respect to $N_2O_5$, use these data to plot a suitable graph.

**(ii)** What is the order of reaction with respect to $N_2O_5$? Explain your answer.

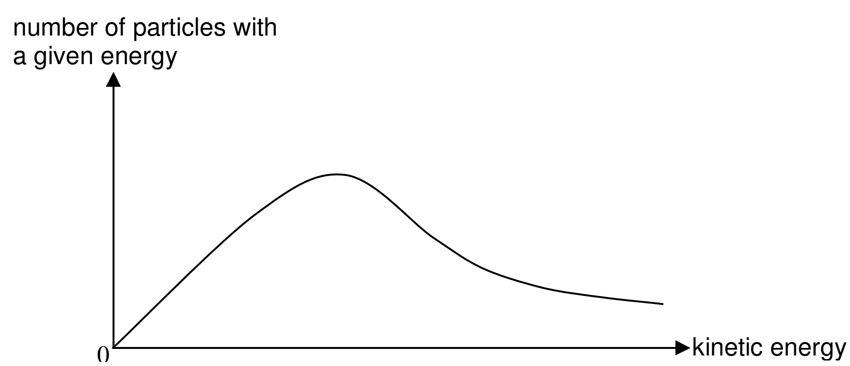
**(iii)** Construct the rate equation for the reaction.

**(iv)** Calculate the rate constant for the reaction, stating its units.

**(v)** Using your answer for **(a)(ii)**, sketch the shape of the graph of $[NO_2]$ against time, assuming $C_o$ to be its concentration when the reaction is complete.

**[9]**

**(b)** The diagram below shows the distribution curve for the energies of hydrogen and nitrogen molecules in an enclosed vessel at 473 K.



**(i)** Copy this diagram on your writing paper. On the same axes, sketch the energy distribution of the same gaseous molecules when the following conditions are applied to the system. Label clearly your curves as **(I)** and **(II)**.

**(I)** The enclosed vessel is cooled to 298 K.

**(II)** More hydrogen and nitrogen gas molecules are introduced into the enclosed vessel at constant temperature.

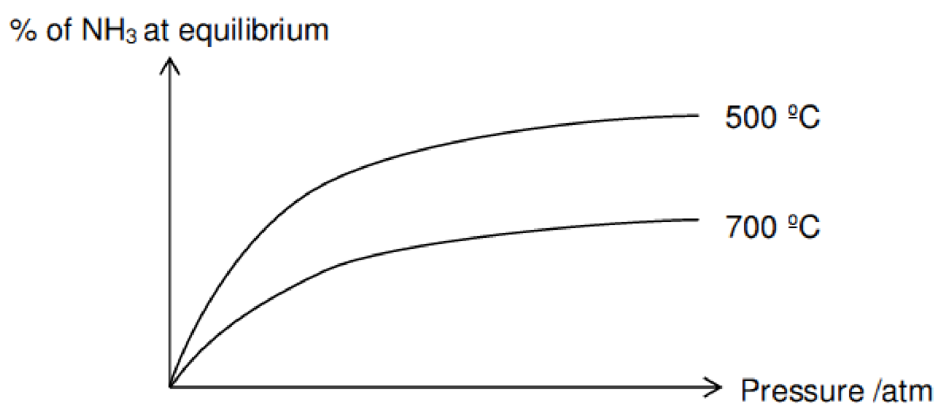
**(ii)** Using collision theory, explain how the condition in **(b)(i)(II)** causes a change in the rate of reaction.

**[5]**

**(c)** In 1908, German chemist, Fritz Haber invented a cheap new source of nitrogen fertilizer, in which he made ammonia, $NH_3$ by capturing nitrogen gas from the air. This process is known as the Haber process.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

The graph below shows the variation of percentage of ammonia in the equilibrium mixture against pressure for the Haber process at 500 °C and 700 °C.




**(i)** Explain the variation of percentage of $NH_3$ with increasing pressure.

**(ii)** Use the above data to deduce whether the production of $NH_3$ is an exothermic or endothermic reaction. Explain your answer.

**(iii)** Describe and explain what would happen to the percentage of $NH_3$ if a catalyst is added.

**[6]**

**[Total: 20]**

**6** **(a)** An ester compound **B** ($M_r$ = 116) has the following composition by mass:
C, 62.0%; H, 10.4%; O, 27.6%.

When **B** is subjected to hydrolysis, an acid **C** and an alcohol **D** are produced.

When 1.00 g of the monoprotic acid **C** is titrated with 0.500 mol dm$^{-3}$ sodium hydroxide, 33.0 cm$^3$ of sodium hydroxide is needed for neutralisation. Alcohol **D** reacts with alkaline aqueous iodine giving a precipitate **E**. When **D** is oxidized to ketone **F**, **F** also reacts with alkaline aqueous iodine, giving the same precipitate **E** and a solution containing a sodium salt **G**.

**(i)** Calculate the molecular formula of **B**.

**(ii)** Suggest suitable reagents and conditions for the hydrolysis of **B**.

**(iii)** Calculate the $M_r$ of acid **C**, and hence identify it.

**(iv)** Describe and explain the reactions of **D** and of **F** with alkaline aqueous iodine. Hence identify the precipitate **E**.

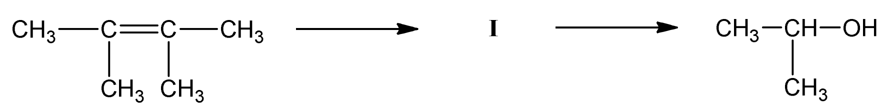
**(v)** Suggest structural formulae of **D**, **F** and **G**.

**(vi)** Hence identify the original ester **B** and write a balanced equation for the hydrolysis.

**[12]**

**(b)** For the two-step syntheses below, identify the reagents and conditions for each step, and write the structures for the intermediate compounds.

**(i)**




**(ii)**


**[6]**

**(c)** The boiling points of the compounds in **(b)(ii)** are given below:

| Compound | Boiling point / °C |
|---|---|
| 2-chlorobutane | 70 |
| 2,3-butanediol | 177 |

Explain the difference in their boiling points, with reference to their structure and bonding.

**[2]**

**[Total: 20]**

**7** The Periodic Table is an arrangement of the elements in increasing order of their atomic numbers, such that elements with similar chemical behaviour are grouped together.

**(a)** The melting points of the higher oxides of the Period 3 elements, sodium to sulfur, are given in the table below.

| **Compound** | **Formula** | **Melting Point / °C** |
|---|---|---|
| Sodium oxide | $Na_2O$ | 1132 |
| Magnesium oxide | $MgO$ | 2830 |
| Aluminium oxide | $Al_2O_3$ | 2054 |
| Silicon dioxide | $SiO_2$ | 1710 |
| Phosphorus pentoxide | $P_4O_{10}$ | 422 |
| Sulfur trioxide | $SO_3$ | 17 |

Explain the differences in melting points, in terms of structure and bonding, between

**(i)** $Na_2O$ and $MgO$

**(ii)** $SiO_2$ and $SO_3$

The higher oxides of the Period 3 elements, sodium to sulfur, can dissolve in or react with water.

**(iii)** Sketch a graph showing the variation in pH of the solution obtained when each of these oxides is added to water.

**(iv)** Explain the shape of the graph, giving balanced equations for any reactions that occur between the oxides and water.

**[9]**

**(b)** An element **R** forms a chloride with the formula **R**C*l*$_x$.

Experiments **1** and **2** are carried out using separate samples of **R**C*l*$_x$.

<u>Experiment **1**</u>
0.45 g of **R**C*l*$_x$ is reacted with an excess of acidified silver nitrate, forming 1.55 g of AgC*l*.

<u>Experiment **2**</u>
0.45 g of **R**C*l*$_x$ is heated strongly, giving off chlorine gas. The resulting residue is then treated with acidified silver nitrate and 0.93 g of AgC*l* is formed.

**(i)** Determine the ratio of the amount of AgC*l* formed in Experiment **1** to the amount of AgC*l* formed in Experiment **2**. Hence, deduce which Group of the Periodic Table element **R** belongs to.

**(ii)** Calculate the $A_r$ of **R** and hence identify **R**C*l*$_x$.

**(iii)** Draw the 'dot-and-cross' diagram of **R**C*l*$_x$ and hence state its molecular geometry.

**[7]**

**(c)** The amount of carbon monoxide present in air can be determined by its reaction with iodine pentoxide, $I_2O_5$, to form carbon dioxide and iodine. The amount of iodine liberated is then determined by titration with a standard solution of sodium thiosulfate.

$$2S_2O_3^{2-} + I_2 \longrightarrow S_4O_6^{2-} + 2I^-$$

**(i)** Write a balanced equation for the reaction between iodine pentoxide and carbon monoxide and identify the type of reaction.

**(ii)** A 100 cm$^3$ sample of polluted air is passed over solid iodine pentoxide and the iodine produced required 15.0 cm$^3$ of 0.10 mol dm$^{-3}$ of sodium thiosulfate for complete reaction.

Determine the concentration (in g dm$^{-3}$) of carbon monoxide present in the sample of polluted air.

**[4]**

**[Total: 20]**

**END OF PAPER**

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